

Automated Printer Troubleshooter

Technical Documentation

Files Covered

PrinterTroubleshooter.ps1 • RunPrinterTroubleshooter.vbs
Version: 1.0 | Platform: Windows | Date: June 2026

1. Overview

The Automated Printer Troubleshooter is a Windows GUI application built in PowerShell that allows IT support staff or end-users to diagnose and resolve common printer problems without using the command line. It features a dark-themed interface with real-time logging, a set of one-click actions, and an optional AI assistant powered by a local Ollama model.

Capability	Detail
GUI Framework	Windows Forms (.NET via PowerShell Add-Type)
Target OS	Windows 10 / 11 (PowerShell 5.1+)
Admin Rights	Required (elevation via VBScript launcher)
AI Integration	Optional — uses local Ollama (llama3 model)
Execution Method	Silent launch via RunPrinterTroubleshooter.vbs

2. File Structure

2.1 RunPrinterTroubleshooter.vbs

A VBScript launcher (3 lines of functional code) whose sole purpose is to start the PowerShell script silently with elevated privileges.

HOW IT WORKS

Uses Shell.Application.ShellExecute with the "runas" verb to trigger a UAC elevation prompt. Passes -NoProfile -ExecutionPolicy Bypass so the script runs regardless of the machine's default execution policy. The console window is hidden (last argument = 1

means SW_SHOWNORMAL for the PS window, but the PS script itself hides it immediately on load).

Key Lines

```
psPath = "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
scriptPath = FSO.GetParentFolderName(WScript.ScriptFullName) &
"\PrinterTroubleshooter.ps1"
shell.ShellExecute psPath, "-NoProfile -ExecutionPolicy Bypass -File ...", "", "runas",
1
```

NOTE

The script path is resolved dynamically relative to the .vbs file location, so both files must be kept in the same folder.

2.2 PrinterTroubleshooter.ps1

The main PowerShell script (~315 lines). It builds the entire GUI using Windows Forms, implements all troubleshooting logic, and optionally calls a local AI model for diagnostic insights.

3. Architecture & Component Layout

The application window (980 × 620 px, fixed size) is divided into two regions:

Region	Size	Contents
Left Panel	280 × 560 px	AI Local Helpdesk Assistant — label + read-only RichTextBox for AI output
Right Area	645 × ~530 px	Printer selector (ComboBox + Refresh button), scrollable log box, progress bar, action buttons, and full diagnostic button

4. Function Reference

4.1 UI Helper Functions

Function	Purpose	Requires Printer
New-StyledButton	Factory: creates a uniformly styled flat button and adds it to the form	No

Function	Purpose	Requires Printer
Show-Loading	Animates the progress bar from 0→100 in 500 ms, then resets to 0	No
Log-Message	Appends a timestamped, color-coded line to the log RichTextBox. Errors also append to \$global:aiContext	No
Ensure-PrinterSelected	Guards actions: logs an error and returns \$false if no printer is selected	No

4.2 Core Troubleshooting Functions

Function	Purpose	Requires Printer
Load-Printers	Queries Get-Printer and populates the ComboBox; logs a prompt to the user	No
Get-PrintQueue	Fetches and logs all active print jobs for the selected printer	Yes
Check-PrinterStatus	Gets printer status via Get-Printer and calls Get-PrintQueue	Yes
Restart-Spooler	Runs Restart-Service Spooler -Force to restart the Windows Print Spooler	No
Clear-PrintQueue	Stops the Spooler, deletes all files in the spool folder, then restarts the Spooler	Yes
Test-Connectivity	Pings the printer's port address; skips test for USB/LPT (local) printers	Yes
Full-Troubleshoot	Runs Check-PrinterStatus → Restart-Spooler → Clear-PrintQueue → Test-Connectivity → Invoke-AIAssistant	Yes

4.3 AI Assistant Function — Invoke-AIAssistant

Invoked automatically at the end of a full diagnostic run. Collects all error messages accumulated in \$global:aiContext and sends them to a local Ollama instance.

- Checks for the ollama command — silently exits if not installed.
- Skips if no errors were collected (\$global:aiContext is empty).
- Builds a structured prompt asking for: Summary, Likely Cause, and Recommended Action (max 3 steps).
- Launches a background job (Start-Job) calling ollama run llama3, piping the prompt via stdin.
- Registers a StateChanged event to receive the result asynchronously and update the AI panel on the UI thread via \$form.Invoke().
- Clears \$global:aiContext after dispatching the job.

DEPENDENCY

Requires Ollama installed and the llama3 model pulled (ollama pull llama3). If Ollama is absent the rest of the tool functions normally — the AI panel simply shows its placeholder text.

5. Global State Variables

Variable	Type	Purpose
\$global:selectedPrinterName	String / \$null	Name of the currently selected printer. Set when the user picks from the ComboBox.
\$global:aiContext	Array of strings	Accumulates error log messages during a session. Consumed by Invoke-AIAssistant and cleared after dispatch.

6. Log Message Color Coding

Type	Color	When Used
"success"	Lime Green	Operation completed successfully (e.g., spooler restarted, queue cleared)
"error"	Indian Red	Failure or problem detected; message also added to \$global:aiContext
"loading"	Sky Blue	In-progress status messages
"info" (default)	White	General informational messages

7. Usage Instructions

7.1 Prerequisites

- Windows 10 or Windows 11
- PowerShell 5.1 or later (included in all supported Windows versions)
- Administrator account or ability to approve a UAC elevation prompt
- (Optional) Ollama installed with the llama3 model for AI diagnostics

7.2 Installation

1. Place both PrinterTroubleshooter.ps1 and RunPrinterTroubleshooter.vbs in the same folder.
2. No installation or registry changes are required.
3. (Optional) To enable AI: install Ollama from <https://ollama.com> and run: ollama pull llama3

7.3 Running the Tool

4. Double-click RunPrinterTroubleshooter.vbs.
5. Accept the UAC elevation prompt when it appears.
6. The Automated Printer Troubleshooter window opens. Printers are loaded automatically.
7. Select a target printer from the dropdown.
8. Click an individual action button or click RUN FULL SYSTEM DIAGNOSTIC.

7.4 Button Actions

Button	Action
Refresh	Reloads the printer list from the system
Check Status	Displays printer status and current print queue
Restart Spooler	Restarts the Windows Print Spooler service (no printer selection needed)
Connectivity	Pings the printer's network port; skips for USB/LPT printers
Clear Queue	Stops the spooler, wipes the spool folder, and restarts the spooler
RUN FULL SYSTEM DIAGNOSTIC	Executes all four actions in sequence, then triggers AI analysis if Ollama is available

8. Known Limitations & Considerations

- The UI is not async — long operations (especially network pings) will momentarily block the form from repainting.
- Clear-PrintQueue stops and restarts the Spooler globally, which will interrupt print jobs on all printers, not just the selected one.
- Test-Connectivity uses the printer's PortName as the ping target. Ports defined by IP address work correctly; named ports (e.g., WSD-xxxxx) may produce false negatives.
- Restart-Spooler does not require a selected printer and will restart the service regardless of ComboBox state.
- The AI model runs in a background job but the result callback updates \$txtAI directly — if the form is closed before the job finishes, the callback will silently fail.
- Execution policy is bypassed at runtime via the -ExecutionPolicy Bypass flag; this does not permanently alter machine policy.

9. Security Notes

⚠ ADMIN REQUIRED

The tool requests elevation via "runas" on every launch. Ensure the .vbs and .ps1 files are stored in a location accessible only to authorized IT staff to prevent misuse.

- Clear-PrintQueue deletes all files under C:\Windows\System32\spool\PRINTERS\ — this is a destructive, irreversible operation.
- -ExecutionPolicy Bypass means unsigned scripts will run. Do not place the script in a shared folder with write access from untrusted users.
- AI prompts are sent to a local Ollama instance only — no data leaves the machine.